

Subset-mass upper-bound score, stated first. At window w , score a candidate right direction $v \in \mathbb{S}^{d-1}$ by

$$\text{score}_w^{\text{UB}}(v) := \frac{\|Bv\|_2^2 + \|A_w v\|_2^2}{1 - \psi_w(v)}, \quad (1)$$

where the leak factor is

$$\psi_w(v) := \sqrt{(n - n_{\text{seen}} - k) n^{-h_2^{(R,w)}(v)/\log(s+k)}}, \quad (2)$$

and $h_2^{(R,w)}(v)$ is the pooled row-response collision entropy computed from the raw rows currently available for entropy estimation,

$$h_2^{(R,w)}(v) := -\log \left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2} \right) = 4 \log \|C_w v\|_2 - 4 \log \|C_w v\|_4, \quad (3)$$

with

$$C_w := \begin{pmatrix} R \\ A_w \end{pmatrix} \in \mathbb{R}^{(s+k) \times d}, \quad C_w v = \begin{pmatrix} Rv \\ A_w v \end{pmatrix}. \quad (4)$$

If this is the first window, take $R = \emptyset$, $s = 0$, $n_{\text{seen}} = 0$, and $B = 0$. If this is a later window with no old row sample, take only $R = \emptyset$ and $s = 0$; the old sketch B and the count n_{seen} still describe the rows processed before the current window. The score is well-defined only when $\psi_w(v) < 1$; see the practical-remarks paragraph for the treatment of the vacuous regime.

Notation in detail. Let $\text{row}(X)$ denote the number of rows of a matrix X . Let

$$A \in \mathbb{R}^{n \times d} \quad (5)$$

be the full data matrix, let A_w denote the current row-window, and let $B \in \mathbb{R}^{r \times d}$ denote the sketch carried from the previous windows. In addition to the sketch, carry a small row sample of old data

$$R \in \mathbb{R}^{s \times d}. \quad (6)$$

Write

$$A_w \in \mathbb{R}^{k \times d}, \quad k = \text{row}(A_w), \quad (7)$$

$$n_{\text{seen}} := \# \text{ raw rows processed before this window}, \quad (8)$$

$$n := \text{total number of rows in the full matrix or stream}, \quad (9)$$

$$v \in \mathbb{S}^{d-1}. \quad (10)$$

Use $[n] := \{1, \dots, n\}$ for row indices. Denote the number of raw rows represented after the current window, the seen-fraction, and the unseen complement-count by

$$m_w := n_{\text{seen}} + k, \quad p_w := \frac{m_w}{n}, \quad q_w := n - m_w = n(1 - p_w). \quad (11)$$

Assume $s+k > 1$ when the pooled entropy proxy is used, and assume $0 < m_w \leq n$.

Explanation of the subset-mass upper bound. This note builds an *upper* estimate for $\|Av\|_2^2$ from a Cauchy–Schwarz subset-mass lemma and a dimension-normalized entropy-preservation heuristic. The only data available at window w are the old sketch B , the current raw window A_w , and the old raw row memory R . The sketch and current window provide an energy estimate, while the raw row memory and current window provide the pooled row-response entropy proxy.

The underlying lemma is purely a statement about a probability distribution

$$p_i := \frac{(Av)_i^2}{\|Av\|_2^2}, \quad \sum_{i \in [n]} p_i = 1, \quad (12)$$

and a subset $S \subseteq [n]$ with $|S| = m_w = n_{\text{seen}} + k$ corresponding to the raw rows processed through the current window. The old sketch summarizes the first n_{seen} processed rows, and the current window contributes the next k raw rows. Under this identification the “seen-row subset mass” is

$$Z_S := \sum_{i \in S} p_i \approx \frac{\|Bv\|_2^2 + \|A_w v\|_2^2}{\|Av\|_2^2} = \frac{E_w(v)}{\|Av\|_2^2}, \quad (13)$$

where $E_w(v) := \|Bv\|_2^2 + \|A_w v\|_2^2$ is the available seen-row energy surrogate. The approximate equality becomes exact when the sketch B preserves the old seen-row response energy on direction v .

Subset-mass lemma. Let $p = (p_1, \dots, p_n)$ satisfy $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$, and let $S \subseteq [n]$ with $|S| = m$. Let $H_2^{\text{full}}(v) := -\log(\sum_{i=1}^n p_i^2) = 4 \log \|Av\|_2 - 4 \log \|Av\|_4$. Then

$$Z_S = \sum_{i \in S} p_i \geq \max \left\{ 0, 1 - \sqrt{(n-m) e^{-H_2^{\text{full}}(v)}} \right\}. \quad (14)$$

Proof of the lemma. Let $S^c = [n] \setminus S$ with $|S^c| = n - m$ and $Z_{S^c} = 1 - Z_S$. Since $p_i^2 \geq 0$, the subset of squared masses lies below the total:

$$\sum_{i \in S^c} p_i^2 \leq \sum_{i \in [n]} p_i^2 = e^{-H_2^{\text{full}}(v)}. \quad (15)$$

By Cauchy–Schwarz applied to the constant-one vector on S^c ,

$$Z_{S^c}^2 = \left(\sum_{i \in S^c} p_i \cdot 1 \right)^2 \leq |S^c| \sum_{i \in S^c} p_i^2 = (n-m) \sum_{i \in S^c} p_i^2. \quad (16)$$

Combining the two displays yields

$$Z_{S^c}^2 \leq (n-m) e^{-H_2^{\text{full}}(v)} \implies Z_{S^c} \leq \sqrt{(n-m) e^{-H_2^{\text{full}}(v)}}. \quad (17)$$

Hence $Z_S = 1 - Z_{S^c} \geq 1 - \sqrt{(n-m) e^{-H_2^{\text{full}}(v)}}$, and the bound $Z_S \geq 0$ is trivial. $\square$

From the lemma to an upper bound on $\|Av\|_2^2$. Applying the lemma with $m = m_w = n_{\text{seen}} + k$ and rearranging $Z_S = E_w(v)/\|Av\|_2^2$ gives, whenever the right side is positive,

$$\|Av\|_2^2 \leq \frac{E_w(v)}{1 - \sqrt{q_w} e^{-H_2^{\text{full}}(v)}}. \quad (18)$$

This bound requires one quantity not directly observable from the window plus sketch: $H_2^{\text{full}}(v)$, the full-matrix collision entropy on direction v .

Dimension-normalized entropy heuristic. Approximate the full-matrix collision entropy by the pooled row-response collision entropy, rescaled by the ratio of the logarithmic support sizes:

$$\frac{H_2^{\text{full}}(v)}{\log n} \approx \frac{h_2^{(R,w)}(v)}{\log(s+k)}. \quad (19)$$

The pooled proxy uses the available rows $C_w = [R; A_w]$ of size $s+k$; the full matrix has n rows. Exponentiating the heuristic,

$$e^{-H_2^{\text{full}}(v)} \approx e^{-h_2^{(R,w)}(v) \log n / \log(s+k)} = n^{-h_2^{(R,w)}(v) / \log(s+k)}. \quad (20)$$

Substituting into the upper bound and collecting constants gives

$$\|Av\|_2^2 \lesssim \frac{E_w(v)}{1 - \psi_w(v)}, \quad \psi_w(v) = \sqrt{q_w n^{-h_2^{(R,w)}(v) / \log(s+k)}}, \quad (21)$$

which is $\text{score}_w^{\text{UB}}(v)$ as declared.

Simplified form. Define the constants

$$q_w := n - n_{\text{seen}} - k, \quad (22)$$

$$\tau_w := \frac{\log n}{\log(s+k)}, \quad (23)$$

$$E_w(v) := \|Bv\|_2^2 + \|A_w v\|_2^2, \quad (24)$$

$$y := C_w v. \quad (25)$$

Then

$$\psi_w(v) = \sqrt{q_w} \exp\left(-\frac{\tau_w}{2} h_2^{(R,w)}(v)\right), \quad (26)$$

and the score factors as

$$\text{score}_w^{\text{UB}}(v) = E_w(v) \gamma_w(v), \quad \gamma_w(v) := \frac{1}{1 - \psi_w(v)}. \quad (27)$$

When $\psi_w(v) < 1$, $\gamma_w(v) > 1$ is a flatness-dependent inflation factor. The inflation factor appears as a pole: small $h_2^{(R,w)}(v)$ drives $\psi_w \rightarrow 1^-$ and $\gamma_w \rightarrow +\infty$.

Exact gradient. Let

$$y_B := Bv, \quad z := A_w v, \quad y := C_w v, \quad (28)$$

and assume $E_w(v) > 0$, $\|y\|_2 > 0$, $\|y\|_4 > 0$, and $\psi_w(v) < 1$. The energy gradient is

$$\nabla E_w(v) = 2B^\top Bv + 2A_w^\top A_w v. \quad (29)$$

The pooled entropy gradient follows by differentiating $4 \log \|C_w v\|_2 - 4 \log \|C_w v\|_4$:

$$\nabla h_2^{(R,w)}(v) = 4 \frac{C_w^\top C_w v}{\|C_w v\|_2^2} - 4 C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right). \quad (30)$$

Here $x^{\odot 3}$ denotes componentwise cubing. For the derivative of ψ_w , assume first that $q_w > 0$. Since $\log \psi_w(v) = \frac{1}{2} \log q_w - \frac{\tau_w}{2} h_2^{(R,w)}(v)$,

$$\nabla \log \psi_w(v) = -\frac{\tau_w}{2} \nabla h_2^{(R,w)}(v), \quad (31)$$

so

$$\nabla \psi_w(v) = \psi_w(v) \nabla \log \psi_w(v) = -\frac{\tau_w \psi_w(v)}{2} \nabla h_2^{(R,w)}(v). \quad (32)$$

The gradient of the denominator log is

$$\nabla \log(1 - \psi_w(v)) = -\frac{\nabla \psi_w(v)}{1 - \psi_w(v)} = \frac{\tau_w \psi_w(v)}{2(1 - \psi_w(v))} \nabla h_2^{(R,w)}(v). \quad (33)$$

Taking logs of the score,

$$\log \text{score}_w^{\text{UB}}(v) = \log E_w(v) - \log(1 - \psi_w(v)), \quad (34)$$

and differentiating,

$$\nabla \log \text{score}_w^{\text{UB}}(v) = \frac{\nabla E_w(v)}{E_w(v)} - \frac{\tau_w \psi_w(v)}{2(1 - \psi_w(v))} \nabla h_2^{(R,w)}(v). \quad (35)$$

Equivalently,

$$\nabla \text{score}_w^{\text{UB}}(v) = \text{score}_w^{\text{UB}}(v) \left[\frac{2B^\top Bv + 2A_w^\top A_w v}{E_w(v)} - 2\kappa_w(v) \left(\frac{C_w^\top C_w v}{\|C_w v\|_2^2} - C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) \right) \right], \quad (36)$$

where the direction-dependent coupling coefficient is

$$\kappa_w(v) := \frac{\tau_w \psi_w(v)}{1 - \psi_w(v)} = \frac{\log n}{\log(s+k)} \cdot \frac{\psi_w(v)}{1 - \psi_w(v)}. \quad (37)$$

The coefficient $\kappa_w(v)$ grows as $\psi_w(v)$ approaches 1, so the entropy-gradient part becomes dominant near the boundary of the non-vacuity region. If $q_w = 0$, then $\psi_w(v) = 0$ for every feasible direction and $\nabla \text{score}_w^{\text{UB}}(v) = \nabla E_w(v)$.

Sign and monotonicity of the entropy term. The entropy contribution to the upper-bound gradient is

$$-2\kappa_w(v) \left(\frac{C_w^\top C_w v}{\|C_w v\|_2^2} - C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) \right) = 2\kappa_w(v) \left(\frac{C_w^\top (C_w v)^{\odot 3}}{\|C_w v\|_4^4} - \frac{C_w^\top C_w v}{\|C_w v\|_2^2} \right), \quad (38)$$

which pushes v toward *lower* $h_2^{(R,w)}(v)$ (since $\kappa_w(v) > 0$ whenever $\psi_w(v) \in (0, 1)$). Thus maximizing this upper-bound score favors higher-leakage directions inside the feasible region: lowering the entropy increases $\psi_w(v)$, enlarges the denominator inflation, and raises the score until the non-vacuity boundary is reached.

Expanded row-memory form. Because

$$C_w^\top C_w v = R^\top R v + A_w^\top A_w v, \quad (39)$$

and

$$C_w^\top (C_w v)^{\odot 3} = R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}, \quad (40)$$

the pooled entropy gradient can be evaluated without explicitly forming C_w :

$$C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) = \frac{R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}}{\|R v\|_4^4 + \|A_w v\|_4^4}, \quad (41)$$

and

$$\frac{C_w^\top C_w v}{\|C_w v\|_2^2} = \frac{R^\top R v + A_w^\top A_w v}{\|R v\|_2^2 + \|A_w v\|_2^2}. \quad (42)$$

Likewise,

$$\psi_w(v) = \sqrt{q_w} \left(\frac{\|R v\|_4^4 + \|A_w v\|_4^4}{(\|R v\|_2^2 + \|A_w v\|_2^2)^2} \right)^{\tau_w/2}. \quad (43)$$

The last identity uses $e^{-h_2^{(R,w)}(v)} = \|C_w v\|_4^4 / \|C_w v\|_2^4$ raised to the power $\tau_w/2$, simplified via the pooled-norm expansions.

Stored-basis form. If

$$B = S V^\top, \quad V = [v_1, \dots, v_r], \quad S = \text{diag}(s_1, \dots, s_r), \quad (44)$$

then

$$\|B v\|_2^2 = \sum_{j=1}^r s_j^2 (v_j^\top v)^2, \quad B^\top B v = \sum_{j=1}^r s_j^2 (v_j^\top v) v_j, \quad (45)$$

so the energy part of the subset-mass gradient is

$$2B^\top Bv + 2A_w^\top A_w v = 2 \sum_{j=1}^r s_j^2 (v_j^\top v) v_j + 2A_w^\top A_w v. \quad (46)$$

The pooled entropy part is independent of this stored-basis representation of B , since B does not appear in C_w .

Optimization problem. To extract a direction, solve

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v=0, \psi_w(v) < 1} \text{score}_w^{\text{UB}}(v), \quad (47)$$

where the columns of Q are the previously accepted directions. The extra $\psi_w(v) < 1$ constraint restricts the feasible set to directions on which the upper bound is non-vacuous. In practice this constraint restricts attention to directions whose pooled collision entropy exceeds the minimum non-vacuity threshold

$$h_2^{(R,w)}(v) > h_{\text{crit}} := \frac{\log q_w}{\tau_w} = \frac{\log q_w \cdot \log(s+k)}{\log n}. \quad (48)$$

Directions with $h_2^{(R,w)}(v) \leq h_{\text{crit}}$ are declared infeasible under this score.

Projected full-gradient ascent. Use the exact gradient of the subset-mass upper-bound score. Project it onto the feasible tangent space:

$$g_{\text{tan}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{UB}}(v). \quad (49)$$

Take

$$\tilde{v}^+ = v + \eta g_{\text{tan}}(v), \quad v^+ = \frac{(I - QQ^\top) \tilde{v}^+}{\|(I - QQ^\top) \tilde{v}^+\|_2}, \quad (50)$$

with $\eta > 0$ chosen by backtracking line search on $\text{score}_w^{\text{UB}}$. Reject any step that leaves the non-vacuity region $\psi_w(v) < 1$.

Basis-expansion interpretation. If one is building a local search subspace rather than taking a pure steepest-ascent step, the new direction added to the basis is the projected full gradient

$$g_{\text{basis}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{UB}}(v). \quad (51)$$

In a reduced-subspace version with current local right search basis $Z \in \mathbb{R}^{d \times q}$, $Z^\top Z = I$, the expansion residual after solving inside $\text{span}(Z)$ is

$$r_Z(v) := (I - ZZ^\top) (I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{UB}}(v). \quad (52)$$

The local basis should be expanded when $\|r_Z(v)\|_2 / \|(I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{UB}}(v)\|_2 > \tau$.

Operator extraction from the retained right subspace. After the scored directions and any auxiliary directions are assembled into an orthonormal right basis $V \in \mathbb{R}^{d \times r}$, form

$$M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}, \quad Y := M_w V, \quad U := \text{orth}(Y), \quad (53)$$

and compress M_w to the left image of the retained right space,

$$\widetilde{M} := U^\top M_w = U_{\text{small}} \Sigma V_{\text{new}}^\top. \quad (54)$$

Here $\text{orth}(Y)$ denotes any matrix with orthonormal columns spanning the column space of Y . Equivalently, $P_U := UU^\top$ is the orthogonal projector onto the left image of the retained right space. The carried sketch is

$$B_{\text{next}} := \Sigma V_{\text{new}}^\top, \quad (55)$$

so the next window receives a compact right-action representation of the compressed operator selected by the optimized basis V .

Why this is the sketch carried into the next window. For any direction v ,

$$\|B_{\text{next}} v\|_2 = \|\Sigma V_{\text{new}}^\top v\|_2 = \|P_U M_w v\|_2, \quad (56)$$

so $\|B_{\text{next}} v\|_2^2$ equals the quadratic old energy on the retained compressed operator. The row memory R_{next} supplies the pooled row-level entropy proxy for the next window's $h_2^{(R, w+1)}$.

First-window special case. If $B = 0$, $R = \emptyset$, $s = 0$, and $n_{\text{seen}} = 0$, then

$$p_1 = \frac{k}{n}, \quad q_1 = n - k, \quad \tau_1 = \frac{\log n}{\log k}, \quad C_1 = A_1, \quad (57)$$

and the first-window score reduces to

$$\text{score}_1^{\text{UB}}(v) = \frac{\|A_1 v\|_2^2}{1 - \psi_1(v)}, \quad \psi_1(v) = \sqrt{n - k} \left(\frac{\|A_1 v\|_4^4}{\|A_1 v\|_2^4} \right)^{\tau_1/2}. \quad (58)$$

The full gradient reduces to

$$\nabla \text{score}_1^{\text{UB}}(v) = \text{score}_1^{\text{UB}}(v) \left[\frac{2A_1^\top A_1 v}{\|A_1 v\|_2^2} - 2\kappa_1(v) \left(\frac{A_1^\top A_1 v}{\|A_1 v\|_2^2} - A_1^\top \left(\frac{(A_1 v)^{\odot 3}}{\|A_1 v\|_4^4} \right) \right) \right], \quad (59)$$

with

$$\kappa_1(v) = \frac{\tau_1 \psi_1(v)}{1 - \psi_1(v)}. \quad (60)$$

Compute retained directions $v_1, \dots, v_r$ sequentially by solving

$$v_j \in \arg \max_{v \in \mathbb{S}^{d-1}, v_i^\top v = 0 \ \forall i < j, \psi_1(v) < 1} \text{score}_1^{\text{UB}}(v). \quad (61)$$

Non-vacuity and the valid direction regime. The score is finite and positive only when $\psi_w(v) < 1$, i.e.

$$h_2^{(R,w)}(v) > h_{\text{crit}} := \frac{\log q_w}{\tau_w}. \quad (62)$$

This threshold statement assumes $q_w > 0$. If $q_w = 0$, all rows have been processed, $\psi_w(v) = 0$, and the subset-mass denominator is exactly 1. For the first window with $n = 128$, $k = 32$, $\tau_1 = \log 128 / \log 32 = 7/5$, $q_1 = 96$:

$$h_{\text{crit}} = \frac{\log 96}{7/5} \approx \frac{4.564}{1.4} \approx 3.26. \quad (63)$$

The first-window maximum pooled entropy is $\log k = \log 32 \approx 3.47$, so the feasible region is narrow and includes essentially only block-flat directions. Spike-like directions (low $h_2^{(R,w)}$) are declared infeasible by this score, implicitly filtering them out rather than scoring them low. As $n_{\text{seen}} + k$ grows, q_w shrinks and the feasible region widens.

Interpretation of the monotonicity flip. The subset-mass upper bound is a statement of the form “the fraction of full-matrix energy captured by the seen rows is at least $1 - \psi_w(v)$ ”: flat directions leak little energy to the unseen complement, while spike directions can leak essentially all of it if the spike lies in the unseen block. Maximizing score $_w^{\text{UB}}(v) = E_w(v)/(1 - \psi_w(v))$ drives v toward directions on which the block energy could plausibly come from a much larger full-matrix energy, i.e. toward higher-leakage (spikier) directions within the feasible region. An objective that instead maximizes a lower estimate obtained by rewarding high pooled entropy would have the opposite preference: it would favor flat directions because their energy is less likely to be hidden in unseen rows.

The upper-bound score is therefore most naturally used either as a *diagnostic* on candidate directions (large score $_w^{\text{UB}}$ means the block evidence is consistent with a large full-matrix energy, but does not imply it) or as a *gate*: the feasibility constraint $\psi_w(v) < 1$ prunes directions on which block evidence cannot support any bounded claim about $\|Av\|_2^2$. One natural streaming use is therefore to optimize a separate lower-estimate score for selecting directions, while using the upper-bound non-vacuity condition as a gate on whether the available block evidence supports a bounded claim.

Practical remarks. The objective is nonconvex in the feasible region, and becomes singular at its boundary $\psi_w(v) = 1$. Steps should be rejected or dampened when $1 - \psi_w(v)$ is numerically tiny; a practical guard is to replace the denominator by $\max(1 - \psi_w(v), \epsilon)$ for small $\epsilon > 0$ during line search, while keeping the feasibility check $\psi_w(v) < 1$ on accepted iterates. When $\|C_w v\|_2$ or $\|C_w v\|_4$ is numerically tiny, regularize by adding $\delta \cdot \|v\|_2^2$ to both $\|C_w v\|_2^2$ and $\|C_w v\|_4^4$, or reject the step.

The expanded gradient form avoids forming C_w explicitly, but the pooled response norms must still include both Rv and $A_w v$.

The row memory should not be interpreted as a replacement for the old sketch. The sketch B supplies retained old energy, while R supplies old raw-row responses for the pooled entropy proxy. A pure reservoir sample gives coverage, but it can miss rare large rows that dominate ℓ_4 terms. A practical memory should therefore combine a uniform reservoir with a small high-response set selected using current sketch directions.

The non-vacuity constraint $\psi_w(v) < 1$ is more restrictive for the first window and loosens as $p_w \rightarrow 1$. A safe implementation caps $\text{score}_w^{\text{UB}}(v)$ at a large finite value when $\psi_w(v)$ approaches 1 from below, and marks the direction as “upper-bound inconclusive” rather than selecting it on the strength of a near-singular denominator.